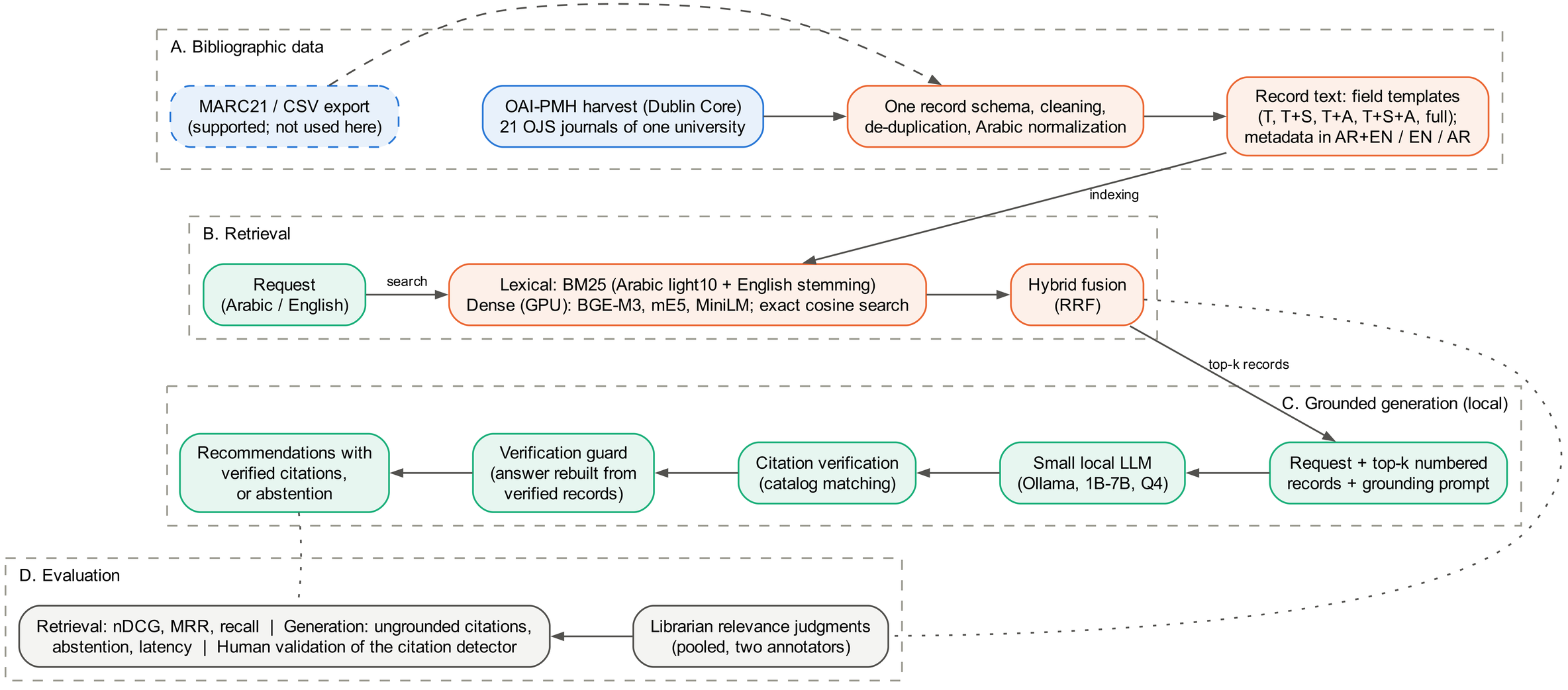
A. Bibliographic data
MARC21 / CSV export (supported; not used here)
OAI-PMH harvest (Dublin Core) 21 OJS journals of one university
One record schema, cleaning, de-duplication, Arabic normalization
Record text: field templates (T, T+S, T+A, T+S+A, full); metadata in AR+EN / EN / AR
indexing
B. Retrieval
Request (Arabic / English)
search
Lexical: BM25 (Arabic light10 + English stemming) Dense (GPU): BGE-M3, mE5, MiniLM; exact cosine search
Hybrid fusion (RRF)
top-k records
C. Grounded generation (local)
Request + top-k numbered records + grounding prompt
Small local LLM (Ollama, 1B-7B, Q4)
Citation verification (catalog matching)
Verification guard (answer rebuilt from verified records)
Recommendations with verified citations, or abstention
D. Evaluation
Librarian relevance judgments (pooled, two annotators)
Retrieval: nDCG, MRR, recall | Generation: ungrounded citations, abstention, latency | Human validation of the citation detector